

DAY 8

SQL Queries for Churn Analysis

Project: Customer Churn Prediction & LTV Engine

Role: Member 3 | Date: 20-05-2026

Objective

Write SQL queries that analyse customer churn patterns from different angles — by contract type, payment method, tenure group, and internet service. These queries will later power dashboards in Superset/Metabase during Week 3.

What is SQL and Why Do We Use It?

SQL (Structured Query Language) is used to ask questions from a database. Instead of loading the whole dataset in Python, SQL lets us filter, group, and summarise data quickly. These queries will connect to our database in Week 3 and feed the dashboards.

File Created

```
sql/queries/churn_feature_queries.sql
```

Key SQL Commands Used

Command	Meaning	Example
SELECT	Choose which columns to show	SELECT Churn, Contract
FROM	Which table to read from	FROM telco_customers
GROUP BY	Group rows by a column value	GROUP BY Contract

COUNT(*)	Count number of rows in each group	COUNT(*) AS total
AVG()	Calculate average of a column	AVG(MonthlyCharges)
ROUND()	Round a number to decimal places	ROUND(AVG(...), 2)
CASE WHEN	Create conditional categories	WHEN tenure <= 12 THEN '0-1 Year'
ORDER BY	Sort results ascending or descending	ORDER BY count DESC

All 6 SQL Queries Explained

Query 1 — Total Churned vs Not Churned

The simplest query. Counts how many customers stayed and how many left overall.

```
SELECT
  Churn,
  COUNT(*) AS total_customers
FROM telco_customers
GROUP BY Churn;
```

Result: Shows 2 rows: one for Churn=Yes, one for Churn=No with counts.

Query 2 — Churn Count by Contract Type

Breaks churn down by contract. Helps answer: do month-to-month customers leave more?

```
SELECT
  Contract,
  Churn,
  COUNT(*) AS customer_count
FROM telco_customers
GROUP BY Contract, Churn
ORDER BY Contract, Churn;
```

Result: Shows churn numbers for each contract type (Month-to-month, One year, Two year).

Query 3 — Average Monthly Charges

Compares average bills of churned vs non-churned customers. Do churners pay more?

```

SELECT
Churn,
ROUND(AVG(MonthlyCharges), 2) AS avg_monthly_charges,
ROUND(MIN(MonthlyCharges), 2) AS min_charges,
ROUND(MAX(MonthlyCharges), 2) AS max_charges
FROM telco_customers
GROUP BY Churn;

```

Result: Shows min, max, and average charges split by churn status.

Query 4 — Customer Count by Payment Method

Shows which payment methods are most common among all customers.

```

SELECT
PaymentMethod,
COUNT(*) AS customer_count
FROM telco_customers
GROUP BY PaymentMethod
ORDER BY customer_count DESC;

```

Result: Returns 4 rows — one for each payment method, sorted by popularity.

Query 5 — Churn Rate by Tenure Group

Groups customers by how long they have been with the company and calculates churn % for each group.

```

SELECT
CASE
WHEN tenure BETWEEN 0 AND 12 THEN '0-1 Year (New)'
WHEN tenure BETWEEN 13 AND 24 THEN '1-2 Years'
WHEN tenure BETWEEN 25 AND 48 THEN '2-4 Years'
ELSE '4+ Years (Loyal)'
END AS tenure_group,
COUNT(*) AS total_customers,
SUM(CASE WHEN Churn='Yes' THEN 1 ELSE 0 END) AS churned,
ROUND(SUM(CASE WHEN Churn='Yes' THEN 1 ELSE 0 END)*100.0/COUNT(*),2) AS
churn_pct
FROM telco_customers
GROUP BY tenure_group
ORDER BY churn_pct DESC;

```

Result: Shows churn percentage for each tenure group. New customers usually have highest churn.

Query 6 — Internet Service vs Churn

Checks which internet service type (DSL, Fiber, None) has the most churners.

```
SELECT
InternetService,
Churn,
COUNT(*) AS customer_count
FROM telco_customers
GROUP BY InternetService, Churn
ORDER BY InternetService, Churn;
```

Result: Returns 6 rows — 2 per service type (Yes/No churn). Fiber optic often shows highest churn.

NOTE: These SQL queries are saved in the .sql file only — they do NOT run yet. They will connect to the actual database in Week 3 when the database is set up.

TIP: SQL comments start with -- (two dashes). Everything after -- on the same line is ignored by the database. Use them to explain what each query does.

Day 8 Deliverables

File	Location	Queries Inside
churn_feature_queries.sql	sql/queries/	6 SQL queries